



DEEP DIVE

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SCRATCH THE SURFACE

KUBERNETES API AND KUBECTL

- REST API including PUT and PATCH with dry-run option
- write your manifests
- kubectl apply vs. replace; patch
 - prefer declarative way with apply for GitOps
 - kubectl patch - partial update using strategic merge, json merge or json patch
 - strategic merge requires merge strategy defined in **API reference**

KUBERNETES CONFIGURATION MANAGEMENT SYSTEMS

- Helm
- Kustomize
- Jsonnet
- Kpt; Ansible; Terraform; Pulumi; etc.

KUSTOMIZE

- native and part of kubectl or standalone binary
- rebase and patch like in software packaging
- embrace manifests
- declarative or imperative
- template free {{ approach }}
- base resource is valid kubernetes manifest
- e.g. configMap key is the smallest item

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KUSTOMIZATION.YAML

kustomize init --auto-detect

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization

namespace: my-cool-app
images: [] # accepts glob
replicas: []
resources: []
patches: []
configMapGenerator: []
secretGenerator: []
```

RESOURCES & COMPONENTS

```
resources:
- base      # directory with kustomization.yaml
- service.yaml
- git@github.com:prometheus-operator/prometheus-operator.git?ref=master # git:: deprecated
- https://github.com/mongodb/mongodb-kubernetes-operator.git//config/default
```

- **Components** like resources but can be included from multiple overlays, transformers (patches) are applied on resources included from different sources

GENERATORS & TRANSFORMERS

- Fundamental concepts - not used directly
- configMapGenerator and secretGenerator - implementations of Generator
- HelmChartInflator is special kind of Generator with **limited support**
- images, namespace, replicas, patches, ... - implementations of Transformer

PATCHES

- choose target using mostly its kind and name or labelSelector
- anchored regex for name and namespace fields
- patch can be a file or inline string

JSON PATCH

- for brief syntax
- Jsonpointer [[RFC-6901](#)]
 - / delimited json path with ~ encoded as ~0, / encoded as ~1
 - cannot address list item by its value
- Jsonpatch [[RFC-6902](#)]
 - add, replace, remove; – append at the end of list

JSON PATCH EXAMPLE

```
- target:
  kind: Deployment
  name: my-(back|front)end
patch: |-
- op: add
  path: /spec/template/spec/volumes/-
  value:
    name: import
    persistentVolumeClaim:
      claimName: import
- op: add
  path: /spec/template/spec/containers/0/volumeMounts/-
  value:
    name: import
    mountPath: /data/import
- op: replace
  path: "/metadata/annotations/haproxy.org~1forwarder-for"
  value: "true"
```

STRATEGIC MERGE

- verbose yet precise
- unknown resource treated as json merge [[RFC-7389](#)]
- can delete whole yaml objects using the special key `$patch: delete`

STRATEGIC MERGE PATCH EXAMPLE

- Delete whole resource

```
- target:
  kind: Service
  name: cilium-agent
  patch: |-
    kind: Service
    metadata:
      name: cilium-agent
    $patch: delete
```

- provide context so **SMP** knows what to change
- metadata.name is required but not used.

REPLACEMENTS

- replace resource value with value from other resource

-	+
the source resource must be part of applied manifest	can address list item by content

multiple targets

partial string replacement

REPLACEMENT EXAMPLE

```
replacements:
- source:
  kind: Namespace
  fieldPath: metadata.name
targets:
- select:
  kind: RoleBinding
  fieldPaths:
  - subjects.[name=etndevel].namespace
```


EXTRAS

- `kubectl diff -k` is noisy

```
kubectl kustomize --enable-helm |  
kubectl diff -f - |  
grep -v -e "f:[[:alnum:]]" -e "k:[^[:alnum:]]" -e "\.: {}" \  
-e "^-    app.kubernetes.io/instance:" -e "^-    labels:"
```

- `kustomize` `localize` resolve remote dependencies
- `helmChart Inflator` (wait for more)
- **KRM functions** little containers that help build final manifests

HELMCHART INFLATOR

- kustomization approach is buggy [#4593](#)
- use generators with manifest of kind: HelmChartInflationGenerator
- use with `--enable-helm`
- local charts directory not **pulled** if present
- generated template can be used as any kustomize base

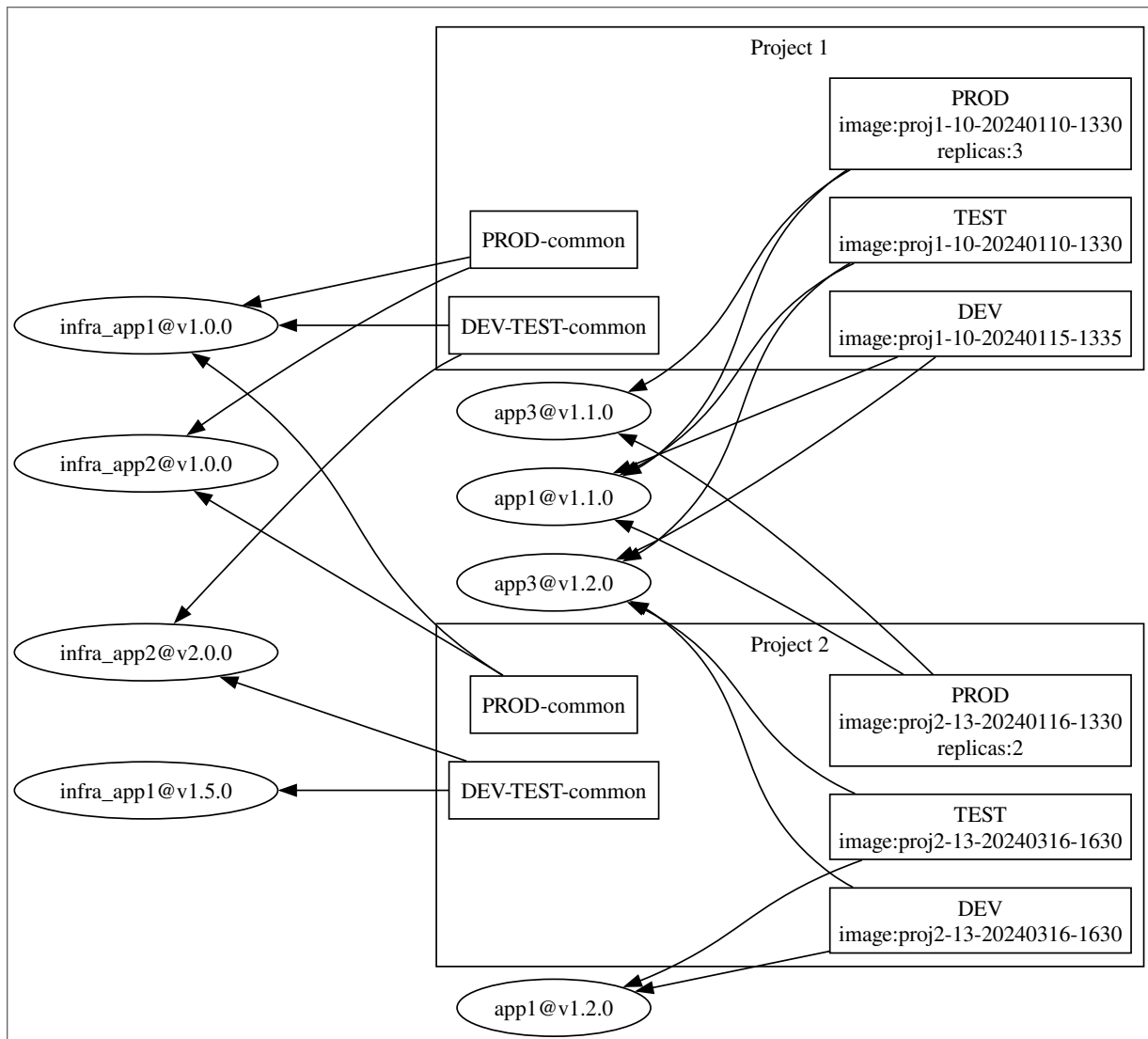


Diagram from the wild

Q&A

*Created with pandoc and reveal.js, from markdown to interactive html;
can be printed to PDF.*

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Speaker notes